



Effect of Segmented Trailing Edge Extensions in Aerodynamic Efficiency

Daniel Curry¹, Sidaard Gunasekaran²

Abstract

The amount and the distribution of vorticity in the wake of AR 4 NACA 0012 wing was affected by passive segmented rigid trailing edge extensions resulting in a decrease in net drag coefficient of the wing by a few percent. Force-based experiments was used to determine the lift coefficient variation with seven segmented trailing edge extensions distributed across the span of the wing. The segmented trailing edge extensions increased the lift coefficient by a few percent due to increased downwash angle observed using streamwise Particle Image Velocimetry (PIV). All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The momentum deficit profiles obtained from PIV were used to determine the drag coefficient of the wing with segmented trailing edge extensions. Investigation of turbulent velocities in the wake of the wing with TE extensions reveal a decrease in RMS and Reynolds stress when compared to the baseline. The turbulent coherent structures behind the TE extensions shows a significant change in the frequency of vortex shedding when compared to the baseline wing as well.

1. Introduction

The parasitic drag contribution on airplanes during cruise is in the order of 50% of the total drag (Butler (1973)). The streamwise pressure gradient created by the periodic shedding of vortices initiates on-body flow separation resulting in higher drag, undesirable structural vibrations, and higher acoustic levels. Different methods are available in the literature to decrease drag of a streamlined/bluff body by affecting the vortex shedding in the wake. A major inconvenience in the aerospace industry today is the fact that most conventional aircraft cannot operate under optimal conditions (maximum Lift-to-Drag ratio) because it is too slow. Hence, a proper solution to reducing induced and parasite drag is still very much needed. The research presented in this paper explores the use of segmented rigid trailing edge extensions as a means to increase lift and decrease drag on a wing. The concept was motivated from the use of long-chain polymers to reduce drag in a pipe flow (Han et al. 2017). The long chain of polymers dampens the turbulent fluctuations and vorticity resulting in drag reduction. Even though this paper only explores the role of a rigid segmented trailing edge extensions in the aerodynamic efficiency, the overall objective of this research is to explore the effect of long segmented flexible trailing edge extensions (similar to the long polymer chains in pipe flow) on the aerodynamic efficiency of the wing.

The use of trailing edge extensions is in no way a new concept in aerodynamic research. Numerous literature exists which explores the use of the trailing edge extensions in a bluff body such as cylinders (Apelt and West (1973), Apelt and West (1975), Karakus et al. (2008)). Cylinders produce oscillating vortices (Von-Karman vortex shedding) in the wake that contribute largely to the drag, pressure fluctuations, sound generation and structural vibration. By attaching a splitter plate perpendicular to the cylinders at the trailing edge, the vortex oscillation is disrupted causing the vortices to disperse. However, the flexible splitter plate proved to be advantageous because it acts as a passive control method to affect vortex shedding. The experiments performed by Teksin and Yayla (2016) on a cylinder with a splitter plate attached showed a decrease of turbulent kinetic energy in the wake of 51 -52% when compared to the

¹ Undergraduate student, Department of Mechanical and Aerospace Engineering, University of Dayton, OH - 45469

² Assistant Professor, Department of Mechanical and Aerospace Engineering, University of Dayton, OH - 45469

cylinder with no splitter plate. The effectiveness of splitter plate did peak when the ratio of plate length and diameter of the cylinder equaled 2.25. The splitter plate is extremely effective in reducing drag behind a bluff body such as cylinders, trucks, square cylinders, etc. but how effective are they in streamlined bodies such as wings? It was actually found to be very effective as well. The reason why a splitter plate (or trailing edge (TE) extensions) are effective in wings because wings also experience vortex shedding at the trailing edge resulting in pressure fluctuations and drag increase. And similar to a cylinder, the vortex shedding behind a wing is also proven to be a function of Reynolds number as shown in experiments done by Yarusevych et al. (2009). They determined that the roll-up of vortices in the separated shear layer play a key role in the flow transition to turbulence. The effect of vortex shedding on the boundary layer can be seen clearly in Figure 1a and 1b taken from Yarusevych et al. (2009) where the smoke released downstream of the wing is seen upstream on top of the wing.

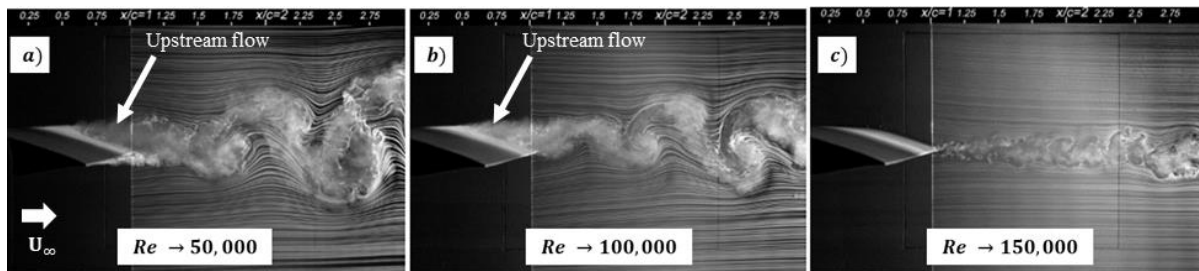


Figure 1 Shedding of vortices from the trailing edge of NACA 0025 airfoil at different Reynolds numbers [Yarusevych et al. (2009)]

Figure 1a and 1b shows prominent turbulent wake vortex shedding due to the separated upper surface shear layer. Huang & Lin (1995) and Huang & Lee (2000) performed experiments on NACA 0012 airfoil and reported that the vortex shedding is only observed at lower Reynolds numbers where boundary layer separation occurs without reattachment. Yarusevych et al. (2009) amended this result and proved that vortex shedding occurs even after boundary layer attaches to the surface at higher Reynolds number as shown in Figure 1c and that the vortex shedding varies linearly with the Reynolds number.

The vortex shedding is also a function of trailing edge geometry. Guan et al. (2016) experimented with multiple beveled trailing edge geometries and showed that even subtle changes in geometry can result in substantial changes in wake signatures. The vortex shedding was found to be greater at the sharp trailing edge when compared to the smooth trailing edges. But even with the smooth trailing edge, the turbulent coherent structures are found to convect without distinct separation points into the wake which complements the result from Yarusevych et al. (2009). Therefore, a NACA 0012 wing was chosen for this study and the effect of the vortex shedding behind the trailing edge, and behind the segmented trailing edge extension was explored.

2.1 Vortex Shedding Mitigation Techniques:

Most of the parasitic drag reduction methods on a wing is targeted at keeping the boundary layer attached and delaying the transition. A slew of active and passive flow control techniques involving laminar flow control, wall cooling, hybrid laminar flow control, active wave suppression, use of riblets, vortex generators, large eddy breakup devices, surface geometry effects such as streamwise and transverse curvatures and microgrooves, synthetic boundary layer, etc. were used to prevent boundary layer transition and separation. But when compared to the number of methods available to mitigate vortex shedding from bluff bodies such as cylinders, trucks, cars, etc. the number of methods available to mitigate vortex shedding from streamlined bodies are minimal.

One of the popular methods to mitigate the influence of vortex shedding from wing is the use of Gurney flaps or divergent trailing edges. Gurney flap is an extension of the trailing edge in the direction perpendicular to the chord. The use of Gurney flap generates a favorable streamwise pressure gradient at high angle of attack and is known to shift the location of the separation from the leading edge to the quarter chord location while at the same time increasing lift on the main airfoil profile (Stanewsky (2001)). A finite pressure differential is carried to the trailing edge and is sustained by a vortex shedding induced base pressure on the downstream face of the flap. Numerous computational and experimental work have been done to study the effect of length and angle of Gurney flaps on vortex shedding. (Neuhart and Pendergraft (1988), Jang and Ross (1998), Storms and Jang (1994), Traub (2017)). But these flaps are usually more effective at higher angles of attack where the flow separates and generates higher drag as expected in areas where flow is not separated. The disadvantage of higher drag using Gurney flaps at lower angles of attack is overcome by employing a static extended trailing edge (SETE) or flexible extended trailing edge (FETE).

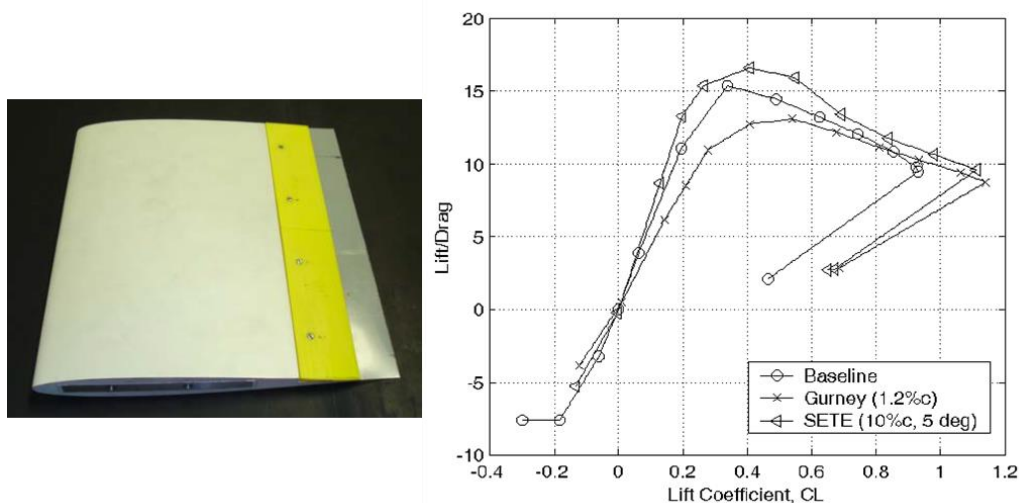


Figure 2 a) NACA 0012 wing with static extended trailing edge. b) Variation of aerodynamic efficiency with coefficient of lift for baseline, Gurney and SETE configurations. The SETE configuration yielded better aerodynamic efficiency than the Gurney flap (Lui et al (2007)).

Lui et al (2007) attached a thin flat plate at the trailing edge of NACA 0012 airfoil made of aluminum and Mylar and determined the changes in the airfoil efficiency as a function of angle of attack and compared it with the measurement made from Gurney flap (Figure 2). SETE showed a larger lift increase at a smaller drag penalty better than a Gurney flap since the SETE was in between the wake of the main airfoil. SETE shows improvement in lift characteristics across the range of angles of attack when compared to Gurney flaps where the lift improvement is seen only at higher angles of attack. Lui et al (2007) also determined the aeroelastic deformation for aluminum (less than 1%) and Mylar (13%) and postulated that MEMS microphones can be embedded in the SETE which will change and react to surroundings. A similar approach is used in this research, but instead of using a full span SETE, the aerodynamics of segmented TE extensions was studied which on comparison to full span SETE, can conserve weight and reduce skin friction drag forces on a wing. Segmented TE extensions can also act as control surfaces which was implemented by Lee and Kroo (2002) where they placed microflaps or Miniature Trailing Edge Effectors (MiTE) on the trailing edge of the high aspect ratio wings (Figure 3) to suppress flutter through dynamic deflection. With this type of controller, they were able to increase the flutter speed by 22%.

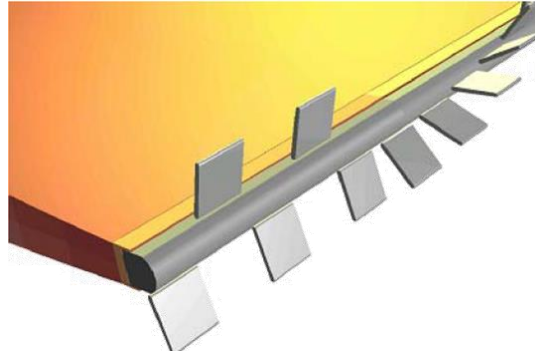


Figure 3 Array of MiTEs (Lee and Kroo (2002))

The literature review indicates that extended trailing edges could be effective in reducing drag and increasing lift in wings. The segmented TE extensions could also lead to drag reduction and control flutter speed and could act as control surfaces. The combination of these different methods of reducing drag display the potential for the segmented TE extensions in this current research. It is hypothesized that the segmented TE extensions will reduce the turbulent fluctuations and vorticity leading to a reduction in drag coefficient.

2. Experimental Setup

2.1 Wind Tunnel

All the experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections. The test section is convertible from a closed jet configuration to an open jet configuration with the freestream range of 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at a freestream turbulence intensity below 0.1% measured by hot-wire anemometer. The tunnel also can vary the freestream velocity profile at up to 5 Hz and over 50% velocity amplitude using a downstream shuttering system. All the experiments mentioned in the paper were done in the open jet configuration where an inlet of 76.2 cm x 76.2 cm opens to a pressure sealed plenum. The effective length of the test section in the open jet configuration is 182cm (72"). A 137cm x 137cm (44" x 44") collector collects the expanded air on its return to the diffuser. A photo of the UD-LSWT open jet configuration is shown in Figure 4. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an Omega differential pressure transducer (Range: 0 – 6.9 kPa).

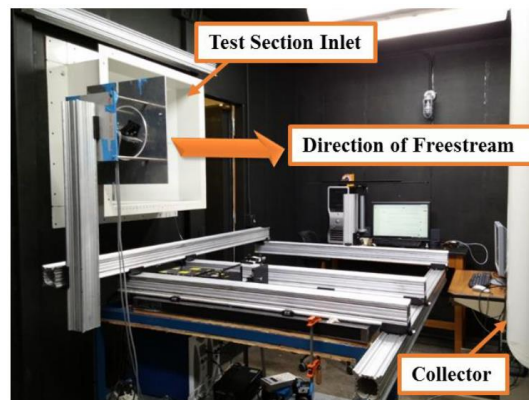


Figure 4 University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open-jet configuration

2.2 Test Model

A NACA 0012 semi-span wing with 20.32 cm span and 10.16 cm chord with capability to attach multiple TE extensions was designed in SolidWorks as seen in Figure 5 and 3D printed using Stratasys uPrint SE Plus printer at the University of Dayton. The model uses two pieces to clamp the TE extensions to the main wing. This allows for high variability of the different configurations of TE extensions. In the research presented in this paper, seven TE extensions made from a plexiglass with a thickness of 1 mm, length of 2.54 cm ($l/c = 0.25$) and a width of 0.635 cm ($d/c = 0.0625$) were used. More variations in the dimensions of the TE extensions and with different stiffness will be tested in a follow up paper. Future testing may incorporate more materials and alternating the length of the streamers along the wing to gain the most effective design. With the trailing edge extensions, the surface area of the wing was increased by 11% when compared to the baseline.

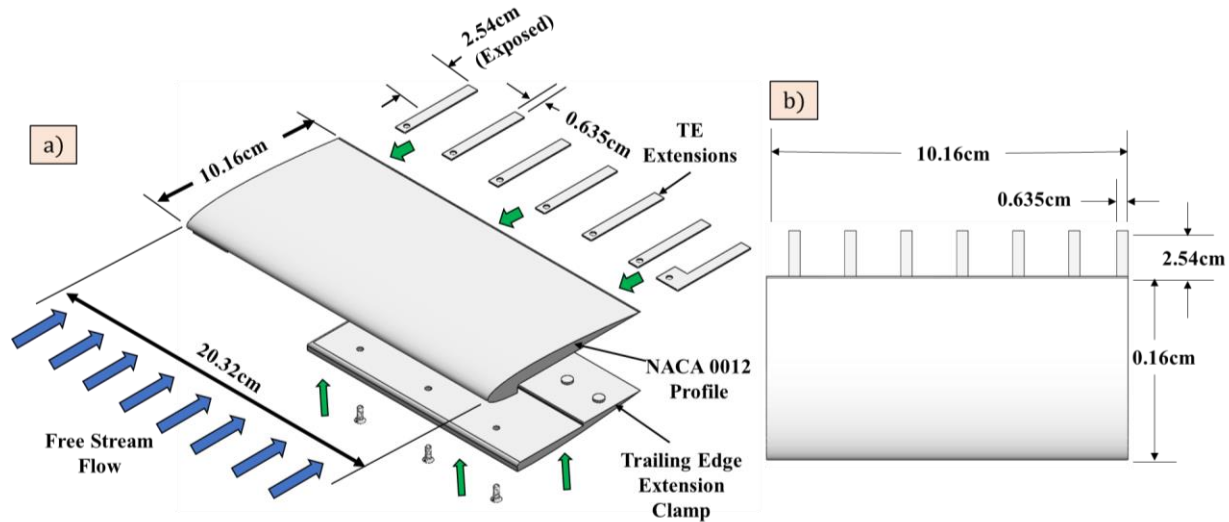


Figure 5 SolidWorks model of semispan AR 2 NACA 0012 wing with streamer inserts a) Exploded view b) Top view of the assembled wing

2.3 Force Based Experiment

Force based experiments were performed on the NACA 0012 semispan model with and without TE extensions at a Reynolds number of 200,000 (Test Matrix shown in Table 1). The models were tested at an angle of attack range from -15° to 15° . Two trials of the same experiment were done with increasing and decreasing the angle of attack to check for hysteresis. The schematic of the force based test setup is shown in Figure 6.

Table 1 Test Matrix for the force based experiments

Force Test Model	Reynolds Number	Angle of Attack (Degrees)	Trials
NACA 0012 (3D)	200,000	-15 to 15	2
NACA 0012 with TE extensions (3D)			

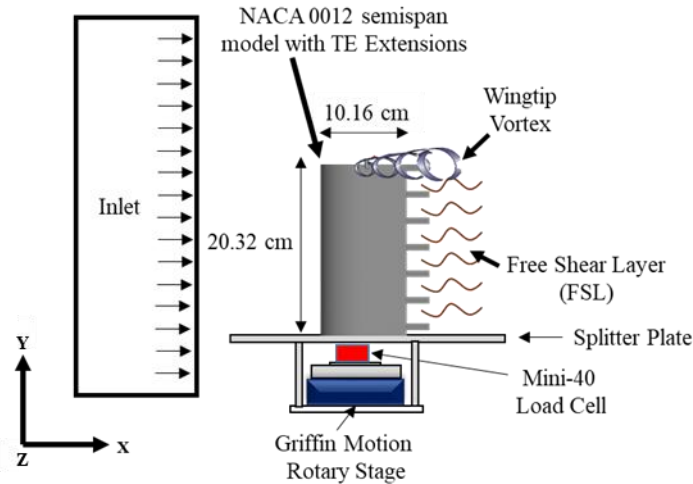


Figure 6 Schematic of the force based experiment test setup for NACA 0012 semispan model with TE extensions. Similar setup was used for NACA 0012 wing without TE extensions as well.

An ATI Mini-40 force transducer was secured underneath the wing at the quarter chord location which interfaced with the Griffin motion rotary stage to change the angle of attack. The rotary stage was controlled using the Galil motion software. The schematic of the test setup is shown in Figure 6. The freestream velocity was measured using a Pitot tube attached to an Omega differential pressure transducer (Range: 0-6.9 kPa).

Force Transducer

An ATI Industrial Automation Mini-40 (www.ati-ia.com) sensor was used to measure the normal and axial forces. Lift and drag forces are obtained from the normal and axial forces using Equation 2 and 3.

$$L = N \cos \alpha - A \sin \alpha \quad (2)$$

$$D = N \sin \alpha + A \cos \alpha \quad (3)$$

The specifications for the Mini – 40 sensor are shown in Table 2. The normal and axial force was measured using the X and Y axis of the sensor with similar magnitude range. The resolution of the forces for all three axes is the same.

Table 2 Specifications of ATI Mini-40

	F _x (N)	F _y (N)	F _z (N)	T _x (Nm)	T _y (Nm)	T _z (Nm)
Range	40	40	120	2	2	2
Resolution	1/100	1/100	1/50	1/4000	1/4000	1/4000

The sampling rate during data acquisition from the Mini-40 was 100 Hz. Tare values were taken before and after each test, and then the average of the two tares are subtracted from the normal and axial force readings.

2.4 PIV Experiment

Streamwise Particle Image Velocimetry (PIV) was conducted in the free shear layer of the NACA 0012 wing wall-to-wall model with and without the TE extensions. Two splitter plates were installed at the wingtips of the wing to prevent the rollup of wingtip vortex and reduce three dimensionalities in the flow.

The PIV measurements were obtained using a Vicount smoke seeder with glycerin oil and a 200 mJ/pulse Nd: YAG frequency doubled laser (Quantel Twins CFR 300). A Cooke Corporation PCO 1600 camera (1600 x 1200 pixel array) with a 105 mm Nikon lens was used to capture the images. One plano-convex lens and one plano-concave lens were used in series to convert the laser beam into a sheet. The laser and the camera were triggered simultaneously by a Quantum composer pulse generator. In each test case, over 700 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. A total of 2 iterations were performed during PIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. Both the streamwise and cross-stream PIV interrogations were conducted at a Reynolds number of 135,000. The test matrix for the PIV experiment is shown in Table 3. The schematic of the PIV test setup is shown in Figure 7a.

Table 3 Test Matrix for Free Shear Layer (FSL) interrogation using PIV

PIV Cases	Wing Model	Reynolds Number	Angle of Attack (Degrees)	Interrogation Location
Streamwise FSL	NACA 0012 wall-to-wall (2D)	135,000	0, 2, 4, 6, 8	Behind TE
	NACA 0012 with TE extensions wall-to-wall (2D)			Behind TE Extension

In the baseline case, the interrogation window was placed near the trailing edge of the wing as shown in Figure 7b to determine the vortex shedding and the momentum deficit. In the wing with the TE extensions, the interrogation window was placed at the trailing edge of the TE extension as shown in Figure 7a and 7c. Nikon 105 mm lens was used in the streamwise PIV case, which gave a spatial resolution of 292 pix/cm in both axes. The size of the field of view was 5.5 cm x 4.1 cm which gave a magnification factor of 0.21. The Δt for the images were set to obtain an average particle displacement of 8-10 pixels in the wake of the wing. The wing model was moved in the spanwise direction to perform PIV behind the hole to maintain the same magnification factor and resolution.

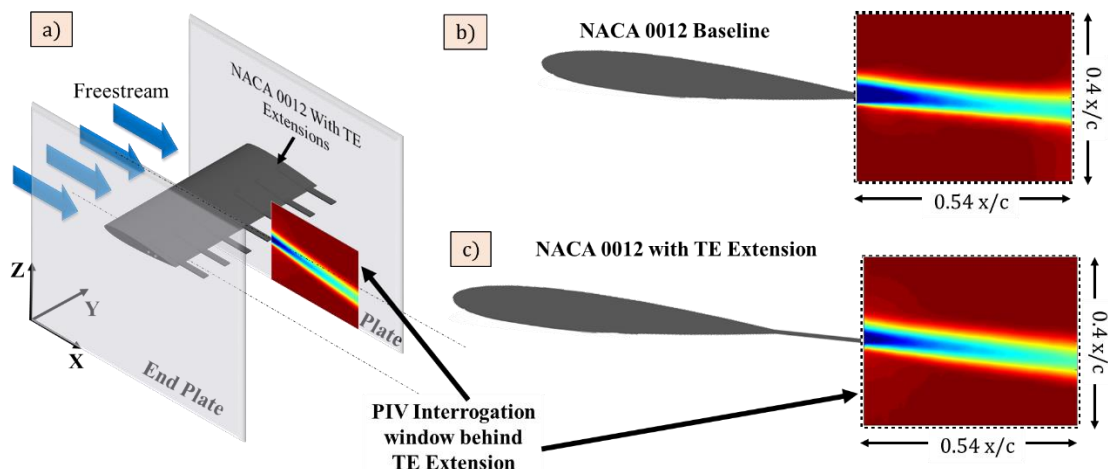


Figure 7 a) Schematic of the PIV test setup for the NACA 0012 wing with TE extensions. Similar setup is used for the baseline wing. The PIV interrogation window for the baseline case was located at the TE (b) and for the wing with TE extension, it was located at the trailing edge of the TE extensions (c)

3. Results

3.1 Force-Based Experimental Results:

The coefficient of lift variation with angle of attack is shown in Figure 8a for the Reynolds number of 200,000 for both the baseline case and the wing with TE extensions. The coefficient of lift variation is compared with the theoretical lift coefficient variation given by McCormick's formula (McCormick (1995)). According to McCormick's formula, the lift curve slope depends on the aspect ratio by,

$$a = \frac{dC_L}{d\alpha} = a_0 \left(\frac{AR}{AR + 2 \frac{(AR + 4)}{(AR + 2)}} \right) \quad (4)$$

where $a_0 = 2\pi$, according to thin airfoil theory. The best fit line of the lift curve gives an effective aspect ratio of 2 which is smaller than the intended aspect ratio of 4. The reduction in effective aspect ratio could be due to the wing-splitter plate interface contributing to three dimensionality of the flow. The baseline results shows good match with the results from Ngo and Barlow (2002) for a Reynolds number of 480,000 for AR 2. The added 11% surface area was taken into account in the calculation of lift from the wing with the TE extensions case. The comparison of lift coefficient magnitude between the baseline and the wing with TE extensions shows almost no variations as a function of angle of attack. Any changes in lift coefficient falls between the uncertainty band of the sensor as indicated by the error bars.

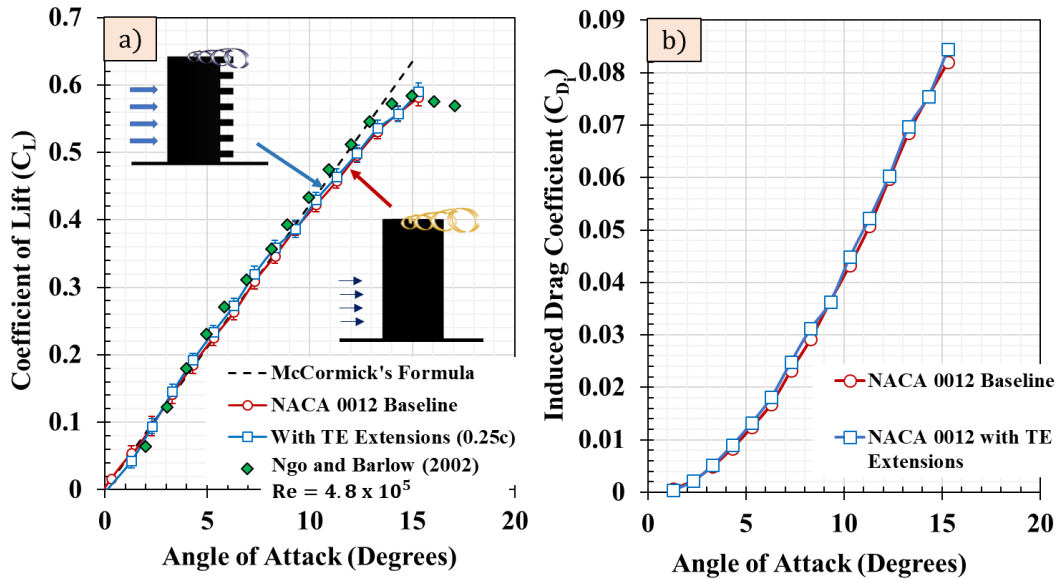


Figure 8 a) Variation of Coefficient of Lift with angle of attack for baseline wing and wing with TE extensions. The lift curve slope shows similar variation with negligible differences between the two cases. b) Variation of coefficient of induced drag for both cases. Since the differences between the

The differences in lift is used to calculate the differences in the induced drag. The induced drag was found by,

$$C_{D_{Induced}} = \frac{C_L^2}{\pi e AR} \quad (5)$$

where e is the span efficiency and AR is the aspect ratio. The span efficiency of the baseline and the 11-hole wing was found using the lift curve slope equation from thin airfoil theory (Equation 6).

$$a = \frac{a_0}{1 + \frac{a_0}{\pi e AR}} \quad (6)$$

Where a is the lift curve slope of the finite wing and $a_0 = 2\pi$. From Equation 6, the span efficiency for the baseline was 0.69 for both the cases since they have the same lift curve slope. Similar magnitude of the induced drag is seen between the baseline and the TE extended case (Figure 8b). Because drag is an order of magnitude less than the lift, the sensor was not capable of measuring the differences in the drag forces between the two cases. Therefore, streamwise Particle Image Velocimetry (PIV) was used to determine the momentum deficit and the parasitic drag of the wing configurations. The results from the PIV are discussed in the section below.

3.2 Momentum Deficit

The streamwise velocity U_x contour obtained behind the TE extension and behind the trailing edge of the baseline wing is shown in Figure 9a. The momentum deficit increases with increase in angle of attack as expected for the both cases. However, subtle differences can be observed in the momentum deficit between the two cases. At a 2° angle of attack, the momentum deficit behind the TE extension is greater than the momentum deficit behind the baseline case. This could be due to increased skin friction drag due to the presence of the TE extension. As the angle of attack increases, the different trend is observed. At 4° and 6° angle of attack cases, the magnitude of the streamwise velocity is almost similar. However, at 8° angle of attack, lower momentum deficit can be observed behind the TE extension when compared to the baseline. This shows that the TE extension was able to reduce the pressure drag of the wing at higher angles of attack. These observations are more apparent in the momentum deficit profiles shown in Figure 8b. The momentum deficit profiles and all the profiles shown in the sections below were taken by averaging 10 data columns in the center of the field of view. The normalized streamwise velocity is plotted against the normalized wake-half width y/L_0 where L_0 is the wake half-width which is considered as the location of 99% U_∞ . The profiles indicate that the momentum deficit at 2° and 4° angle of attack between the two cases are similar. At 6° and 8° angle of attack however, clear differences between the two cases can be seen. The momentum deficit behind the TE extension is clearly lower than the baseline. The momentum deficit profiles shown in Figure 8b was used to determine the total parasitic drag coefficient of the baseline NACA 0012 wing and the wing with TE extensions. The total parasitic drag coefficient of the wing was found by integrating the sectional drag coefficient along with wingspan. The sectional drag coefficient behind the trailing edge and the hole was determined using Equation 7.

$$C_D = \frac{\rho U_\infty^2}{q_\infty S} \int \frac{U_x}{U_\infty} \left(1 - \frac{U_x}{U_\infty}\right) dy \quad (7)$$

The drag coefficient variation with angle of attack is shown in Figure 10. As expected, the drag coefficient behind the trailing edge varies non-linearly with the angle of attack for both cases. The magnitude of the drag coefficient behind the TE extension is lower than the baseline at all angles of attack. Even though the momentum deficit profiles in Figure 9b look similar for both the cases at lower angles of attack, the TE extension case has a higher chord length and surface area when compared to the baseline case which lead to higher drag coefficient. Therefore, for a similar normalized momentum deficit profile, the coefficient of drag is lower in the wake behind the TE extension.

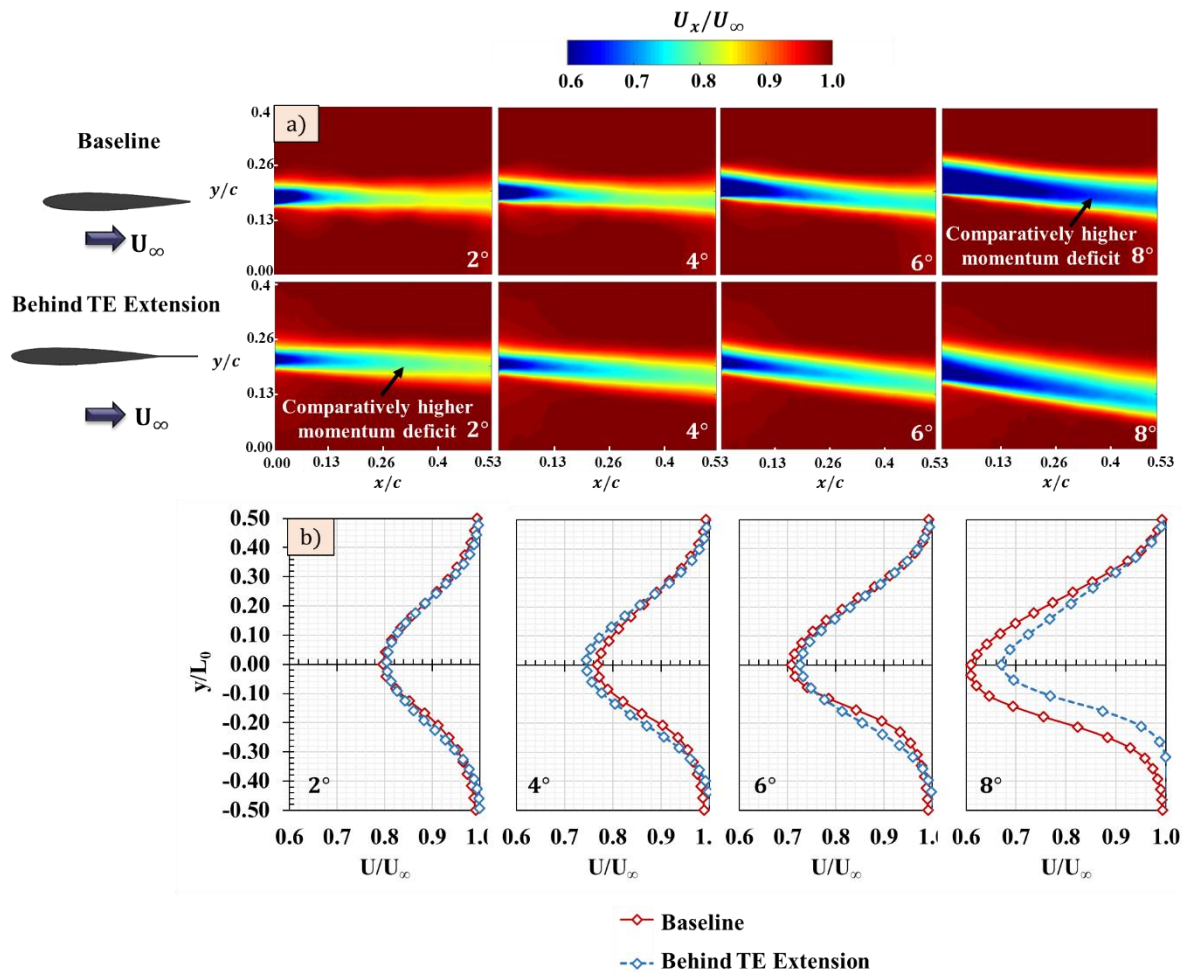


Figure 9 a) Streamwise velocity contours in FSL behind the trailing edge of the NACA 0012 wing and behind the TE extension b) Momentum deficit profiles at different angles of attack for both cases. The momentum deficit behind the TE extension is lower at higher angles of attack when compared to the baseline.

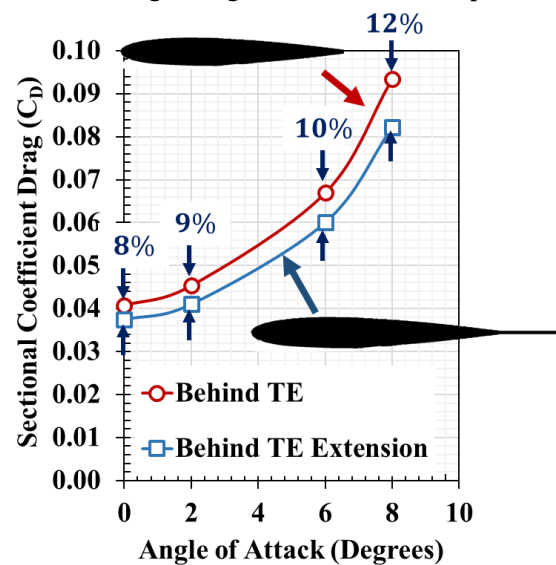


Figure 10 Variation of coefficient drag estimation behind the trailing edge and TE extension a function of angle of attack. The estimated drag coefficient behind the TE extension is lower than the drag coefficient behind the hole.

The differences in the sectional coefficient of drag increases with increase in angle of attack from 8% at 2° to 12% at 8°. To determine the total parasitic drag coefficient of the entire wing with and without TE extensions, a theoretical distribution of the sectional drag coefficient is plotted in Figure 11 for 8-hole wing using the drag coefficient values obtained from the momentum integral shown in Figure 10. The sections of the wing with the TE extension has a lower drag coefficient than trailing edge as observed in Figure 10. The theoretical drag coefficient distribution increases with angle of attack as expected.

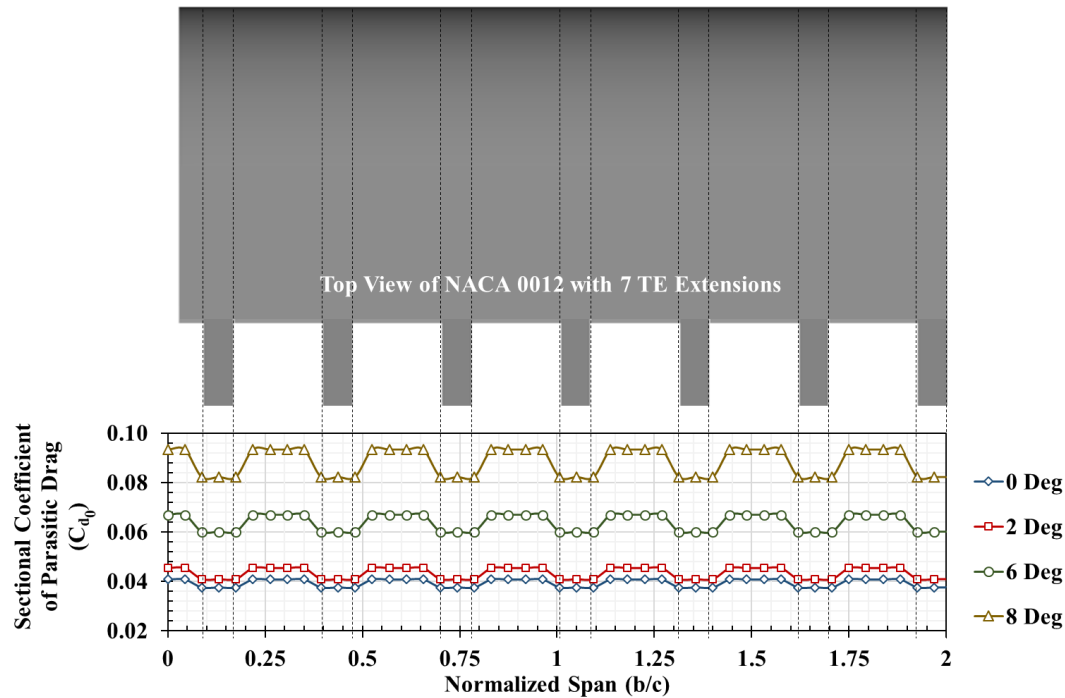


Figure 11 Section drag coefficient variation across the span for wing with seven TE extensions. The drag coefficient behind the TE extension is lower than the drag coefficient behind the trailing edge of NACA 0012.

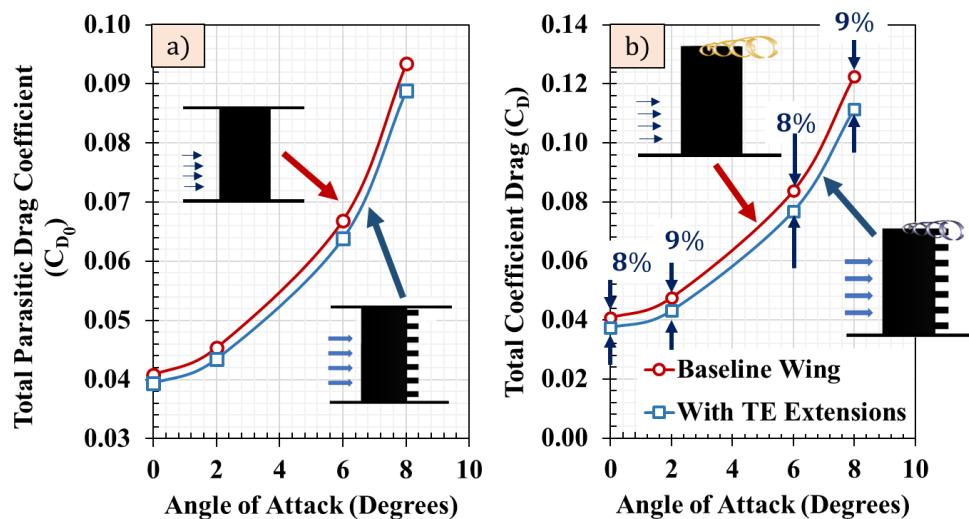


Figure 12 Variation of a) net parasitic drag coefficient and b) total drag coefficient of the baseline wing and wing with TE extensions. The wing with TE extensions show an average decrease in drag coefficient around 8%.

The total drag coefficient of the wing was found by integrating the sectional drag coefficient along the span of the wing. The net parasitic drag coefficient of the NACA 0012 baseline wing and the wing with TE

extensions are shown in Figure 12a for multiple angles of attack. The total parasite drag of the wing with TE extensions is lower than the NACA 0012 baseline wing across all angles of attack. Adding the induced drag found from force based experiment and parasitic drag data found from PIV, the total drag coefficient for the baseline and the wing with TE extension is shown in Figure 12b. The total drag for the wing with TE extensions is also lower when compared to the baseline at all angles of attack. Since the induced drag remained the same for both the cases (Figure 8b), the total drag shows the same trend as the parasite drag coefficient. Total drag reduction in the order of 8% is observed at an angle of attack of 0° increasing to 9% at 8° angle of attack. The average reduction in drag coefficient due to the TE extensions is around 8%. This result indicates that the TE extensions are effective at all angles of attack.

3.3 Y-Vorticity:

The vorticity contours and profiles behind the wake of the baseline NACA 0012 wing and with the TE extensions can be seen in Figure 13a and 13b. The Y-vorticity in the wake was determined by

$$\omega_Y = \left(\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} \right) \quad (8)$$

The velocity gradients in Equation 8 were determined by central difference technique using the experimental velocity data.

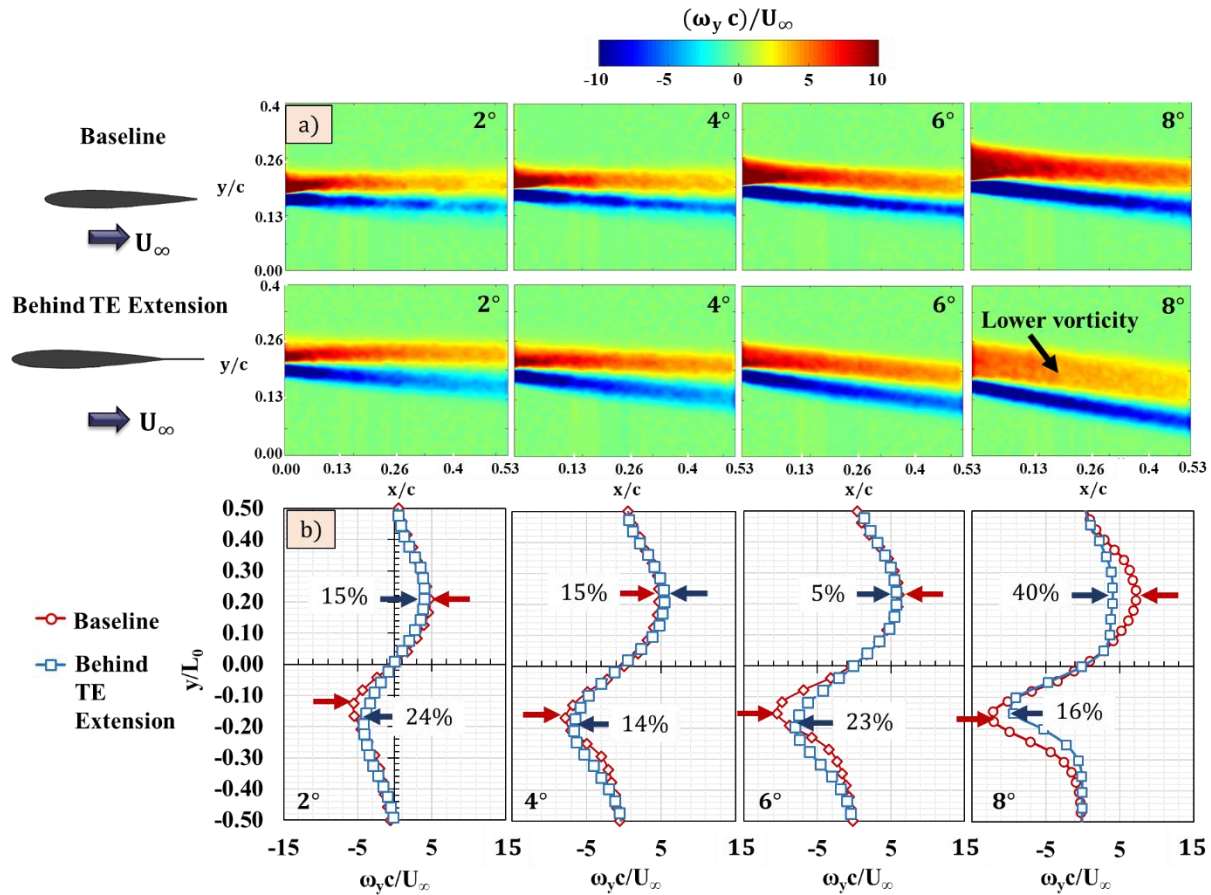


Figure 13 a) Streamwise vorticity contours in FSL behind the trailing edge of the NACA 0012 wing and behind the TE extension b) Vorticity profiles at different angles of attack for both cases. The local vorticity behind the TE extension is lower at higher and lower angles of attack when compared to the baseline.

Similar magnitudes between the two cases is observed in the vorticity contours at lower angles of attack. However, at 8° angle of attack, higher vorticity magnitude is observed behind the TE extension. The vorticity profiles across the contours are compared between the two cases in Figure 13b. At a 2° angle of attack, a reduction in vorticity of 15% on the top surface and 24% on the bottom surface was observed with TE extensions. Then at 4°, the difference in peak vorticity dipped, and the top surface had a 15% increase, and the bottom had a 14% decrease with TE extensions. However, at 8° angle of attack, the extensions reduce the peak vorticity strength by 40% in the top surface and 16% on the bottom. This shows that the TE extensions are most effective at higher angles of attack as the pressure drag begins to dominate over skin friction drag. The reduced vorticity at 8° angle of attack also indicates changes in the vortex shedding frequency and magnitude which is discussed in the next section. The reduction in vorticity also indicates a reduction in total pressure loss through Crocco's theorem which in turn reduces drag.

3.4 Coherent Structures in Wake

As seen in the literature review section, vortex shedding frequency and turbulent length scales contributes to turbulence-induced pressure fluctuations, sound generation, and structural vibrations. The effect of the TE extensions on vortex shedding frequency and turbulent length scales can be determined by comparing the changes in the coherent structures present in the wake between the baseline wing and wing with TE extensions. The coherent structures can be determined by performing two-point correlation of fluctuating velocities (u' and v') in the wake. The two-point correlation also allows to determine the length scales associated with the coherent turbulent motions. Bendat and Piersol (1986) defined the two-point correlation as

$$\rho_{u_i u_j}(X_1, X_2, \tau) = \frac{[u'_i(X_1, t) * u'_j(X_2, t + \tau)]}{\sqrt{u'_i(X_1)^2} \sqrt{u'_j(X_2)^2}} \quad (9)$$

where X_1 and X_2 are two spatial locations in the PIV field of view, τ is the time delay (which is chosen to be zero for the results shown below), and u' represents the fluctuating velocities in i and j direction. Figure 14 and Figure 15 show the contour levels of the normalized two-point correlation functions with zero time delay of the streamwise (u) and transverse (v) fluctuating velocities respectively for angles of attack of 2°, 4°, 6° and 8° angle of attack in the wake of the baseline wing and in the wake of the TE extension. In each case, the reference point (X_1) is chosen to be at the center of the upper shear layer as indicated in Figure 15. The intent is to highlight the correlation in velocity fluctuations between the boundary layer flow from the upper surface and the near wake. The ρ_{uu} contour images for the baseline case shows extensive coherent structures of alternating positive and negative correlation values. Specifically, spatially alternating regions of positive and negative correlation are indicative of the spatially and temporally periodic motions of the fluid. These motions can be related to the tonal character of fluctuations in the flowfield at the frequency of vortex shedding. In the baseline case, the coherent structures are well formed in the shear layer emanating from the lower surface of the wing. The magnitude of correlations in the lower surface shear layer is also higher when compared to the upper surface shear layer. As the angle of attack increases, the length scales (represented by the horizontal distance of each coherent structure) decreases due to increased vortex shedding frequency. This can be observed by quantifying the number of coherent structures in the wake. At 2° angle of attack, there are eight coherent structure and at 8° angle of attack, there are ten coherent structures. However, in the wake of TE extension, the correlation of the upper surface shear layer and the near wake is significantly lower when compared to the baseline case. This indicates a comparatively weaker vortex shedding and turbulent fluctuations in the wake of the TE extension. It is interesting to note that with the increase in the downstream distance, the correlation of the velocity fluctuations in the TE extension

case almost goes to zero at all angles of attack. However, in the baseline case, there is a strong correlation across the field of view.

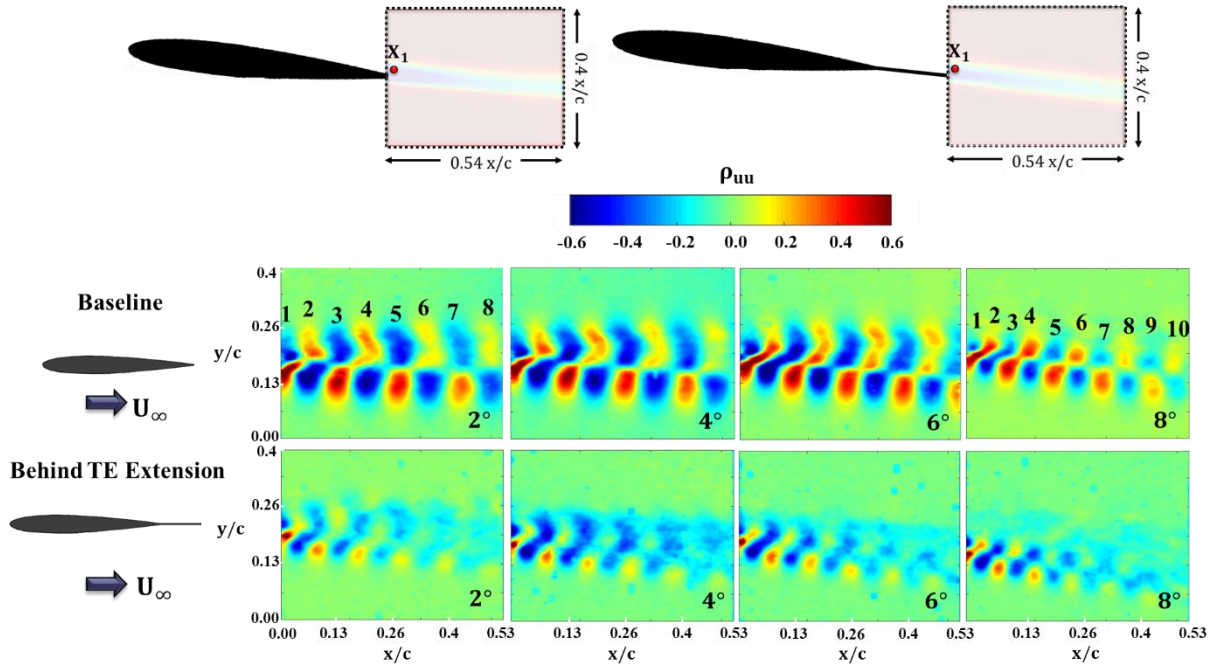


Figure 14 Contours of two-point correlation of the streamwise velocity component for the baseline wing and the wing with TE extensions. Weaker correlations are observed in the wake behind the TE extension indicating lower length scales and velocity fluctuations.

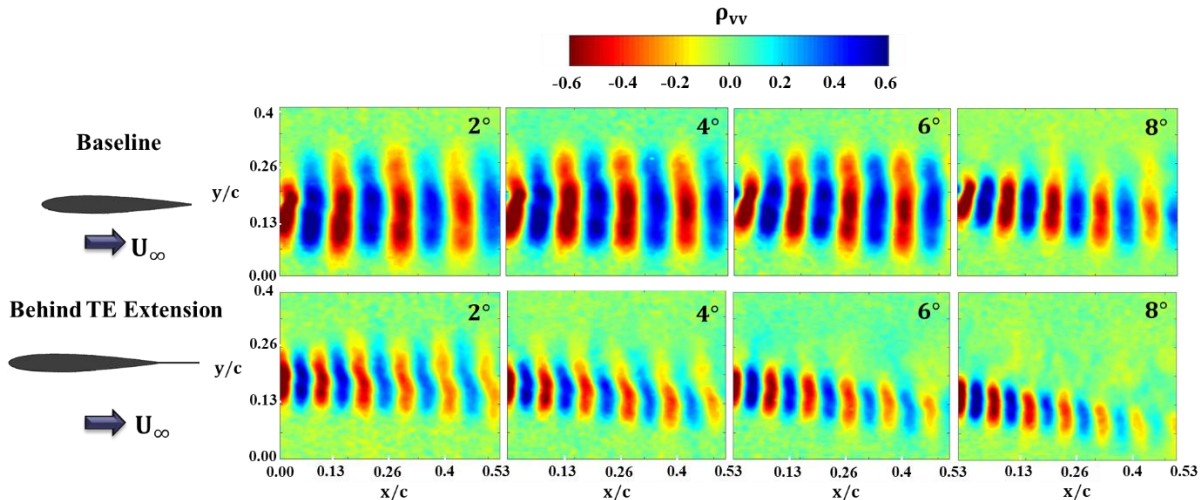


Figure 15 Contours of two-point correlation of the transverse velocity component for the baseline wing and the wing with TE extensions. Similar to ρ_{uu} , weaker correlations are observed in the wake behind the TE extension lower velocity fluctuations. Also, the decrease in wavelength of the correlation indicates a decrease in turbulent length scales.

Similar behavior is observed in the ρ_{vv} (transverse) velocity correlations in the wake of the baseline wing and in the wake of the TE extension. The large alternating regions of positive and negative correlation in v appear in the baseline and the TE extension cases but the wavelength of the correlations in the TE extension is almost of half of that of the baseline wing. The reduction in length scales indicate lower velocity fluctuations in the wake of the TE extension which results in lower pressure fluctuations and lower drag as observed in Figure 12b. The decrease in turbulent length scales decreases the energy cascade in the

turbulence spectrum and increases the viscous dissipation which results in lower drag coefficient of the wing. The reduced fluctuations in the TE extension case can also be seen in the RMS quantities of the streamwise and transverse velocities.

3.4 Reynolds Stress

The Reynolds stress components are indicative of the turbulent intensity within a developing shear layer. Mohsen (1968) suggested that the local maximum Reynolds stress $(u'v')_{max}$ in the Reynolds stress profile may be correlated to the large pressure fluctuations. Therefore, the Reynolds stress distribution in the wake are of great interest as they can indicate how TE extensions affect the amount of turbulence in the flow. The contour plots and profiles of the Reynolds stress comparing the NACA 0012 baseline wing and with TE extensions can be seen in Figure 16a and 16b.

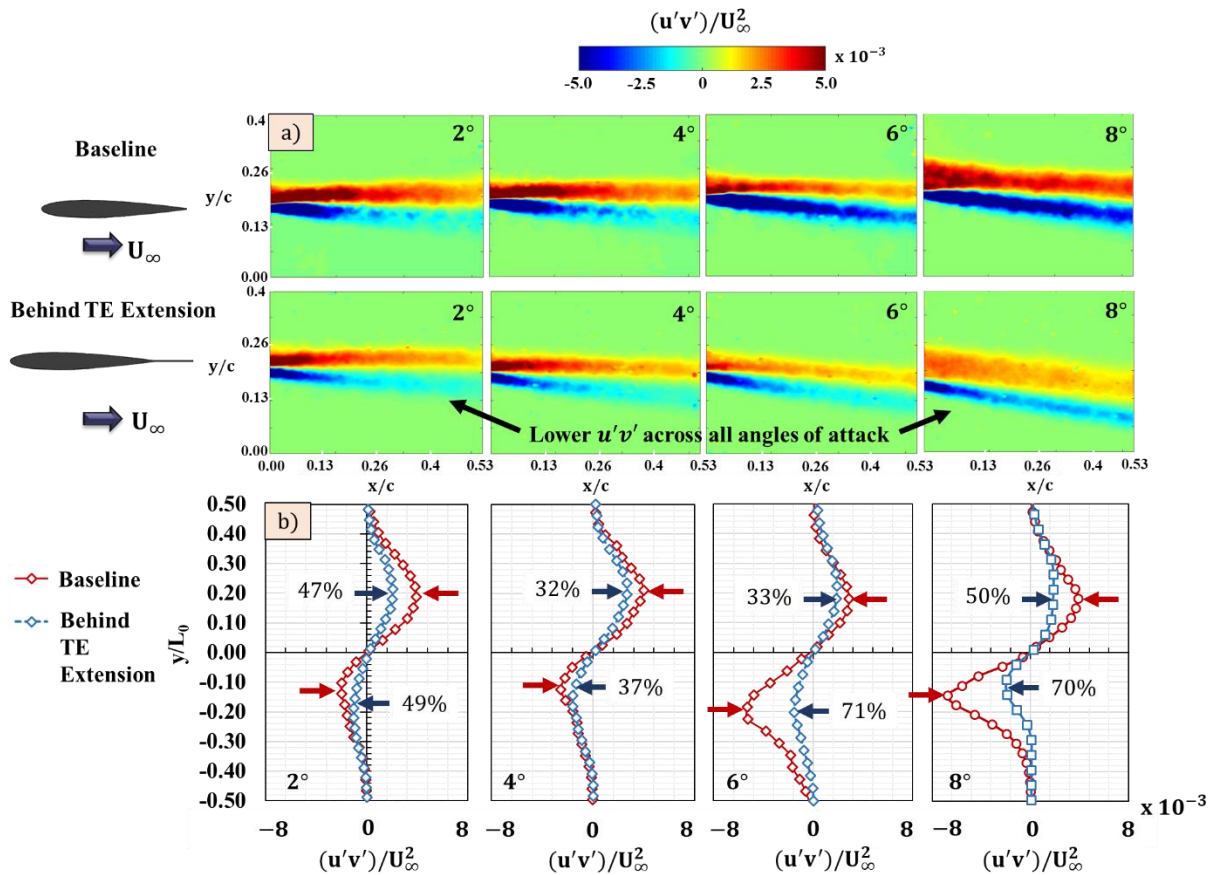


Figure 16 a) Streamwise Reynolds stress contours in FSL behind the trailing edge of the NACA 0012 wing and behind the TE extension b) Reynolds stress profiles at different angles of attack for both cases. The Reynolds stress behind the TE extension is lower across all angles of attack when compared to the baseline.

The magnitude of the both the upper and lower surface Reynolds stress behind the TE extension is lower than the baseline case at all angles of attack. The lower magnitude behind the TE extension indicates that the turbulent fluctuations emanating from the upper and lower surface boundary layer has reduced drastically when compared to the baseline. In both the cases, the Reynolds stress varies in the streamwise direction. But the changes in the streamwise direction in the Reynolds stress is greater in the baseline case when compared to the TE extension case. This trend can be seen at 8° angle of attack. A uniform variation in the Reynolds stress can be observed behind the extended TE case where the Reynolds stress decreases with increase in downstream distance behind the trailing edge of the baseline wing. The differences in the

Reynolds stress between the two cases can be seen clearly in the profiles shown in Figure 16b. Surprisingly, the magnitude of the Reynolds stress in the upper surface is lower than the magnitude of the Reynolds stress in the lower surface in both cases. But the magnitude of the Reynolds stress in both the upper and lower surface of the TE extension case is lower than the baseline in all angles of attack. The Reynolds stress is lowered by 47% and 49% on the upper and lower surfaces respectively at 2° with the TE extensions. The peak stress difference then decreases to 32% and 37% at 4° angle of attack with extensions and again greatly increase at higher angles of attack. This reduction in effectiveness displays a trend similar to the vorticity profiles. At 8° the peak Reynolds stress with the TE extensions is 50% lower on the upper surface and 70% on the lower surfaces. It is interesting to note that the TE extension affects the Reynolds stress in the lower surface significantly more than the upper surface.

3.5 Root-mean-square velocities:

The root mean square of U and Y velocity was determined by

$$U_{RMS} = \sqrt{u_X'^2} \quad Y_{RMS} = \sqrt{u_Y'^2} \quad (10)$$

where u_X' is the fluctuating velocity about the x-axis, u_Y' is the fluctuating velocity about the y-axis. The freestream normalized U_{RMS} is shown in Figure 17 for both the baseline and the wing with TE extension. The magnitude of U_{RMS} increases with increase in angle of attack for both cases but, in all the angles of attack, the magnitude of the RMS is greater in the wake behind the baseline wing when compared to the TE extension. Therefore, the fluctuations in the U velocity are reduced significantly by the TE extension.

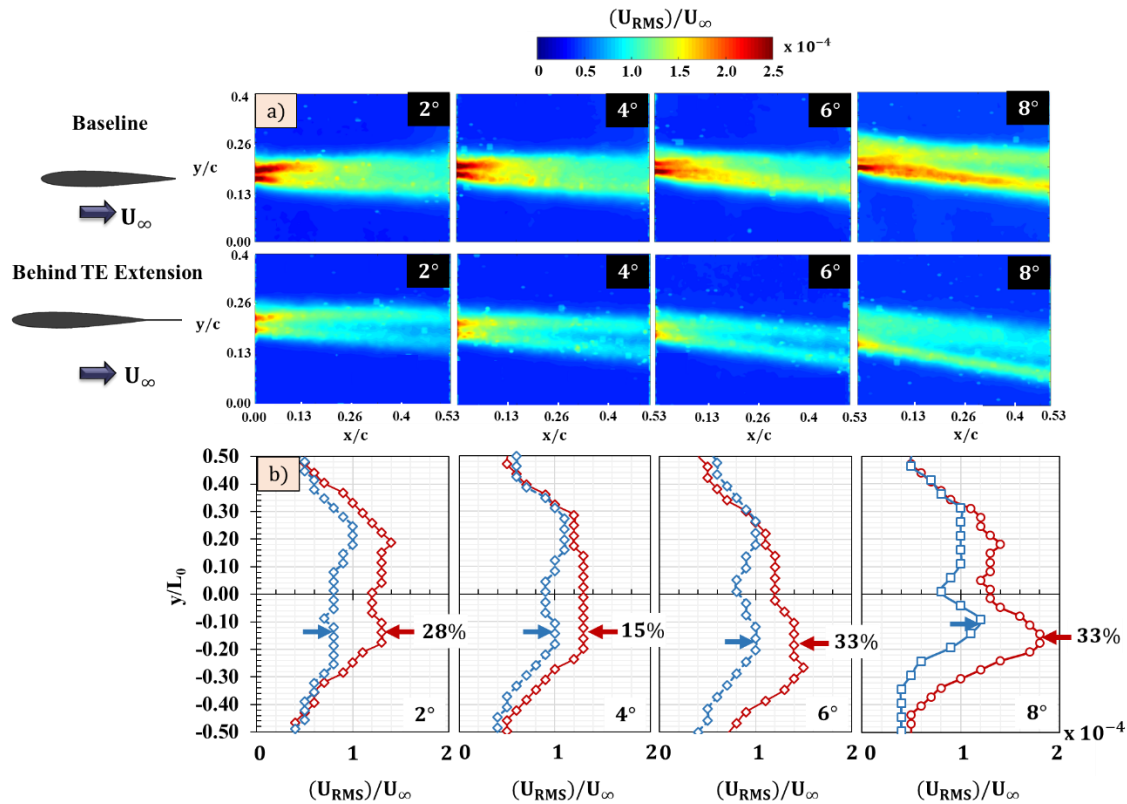


Figure 17 a) Streamwise U_{RMS} contours in FSL behind the trailing edge of the NACA 0012 wing and behind the TE extension b) U_{RMS} profiles at different angles of attack for both cases. Large decreases were observed in the U_{RMS} of the TE extension

In both cases, the U_{RMS} in the lower surface was found to be greater than the upper surface across all angles of attack. This result compares well with the increase Reynolds stress in the lower surface when compared to the upper surface of the wing. The differences in the U_{RMS} can be seen clearly in the U_{RMS} profiles shown in Figure 17b. The average difference in the peak U_{RMS} behind the baseline wing and behind the TE extension is around 30%. Similar results are observed in V_{RMS} as well which is shown in Figure 18a. The normalized V_{RMS} values of 1.8×10^{-5} and 2.0×10^{-5} are highlighted in white to distinctly observe the free shear layer. Similar to the U_{RMS} , the magnitude of the V_{RMS} in the wake behind the TE extension is significantly reduced. The differences can also be clearly seen in the V_{RMS} profiles shown in Figure 18b. The peak differences in V_{RMS} profiles display consistent reductions with TE extensions at an average of 57%. The magnitude decrease in V_{RMS} is greater than the magnitude decrease in U_{RMS} in the wake behind the TE extension. Similar trend was seen in the coherent structures. The two-point correlation of the transverse velocity showed significant changes when compared to the streamwise velocity correlation. The reduction in the length scales might be the cause of lower fluctuations in the wake behind the TE extension.

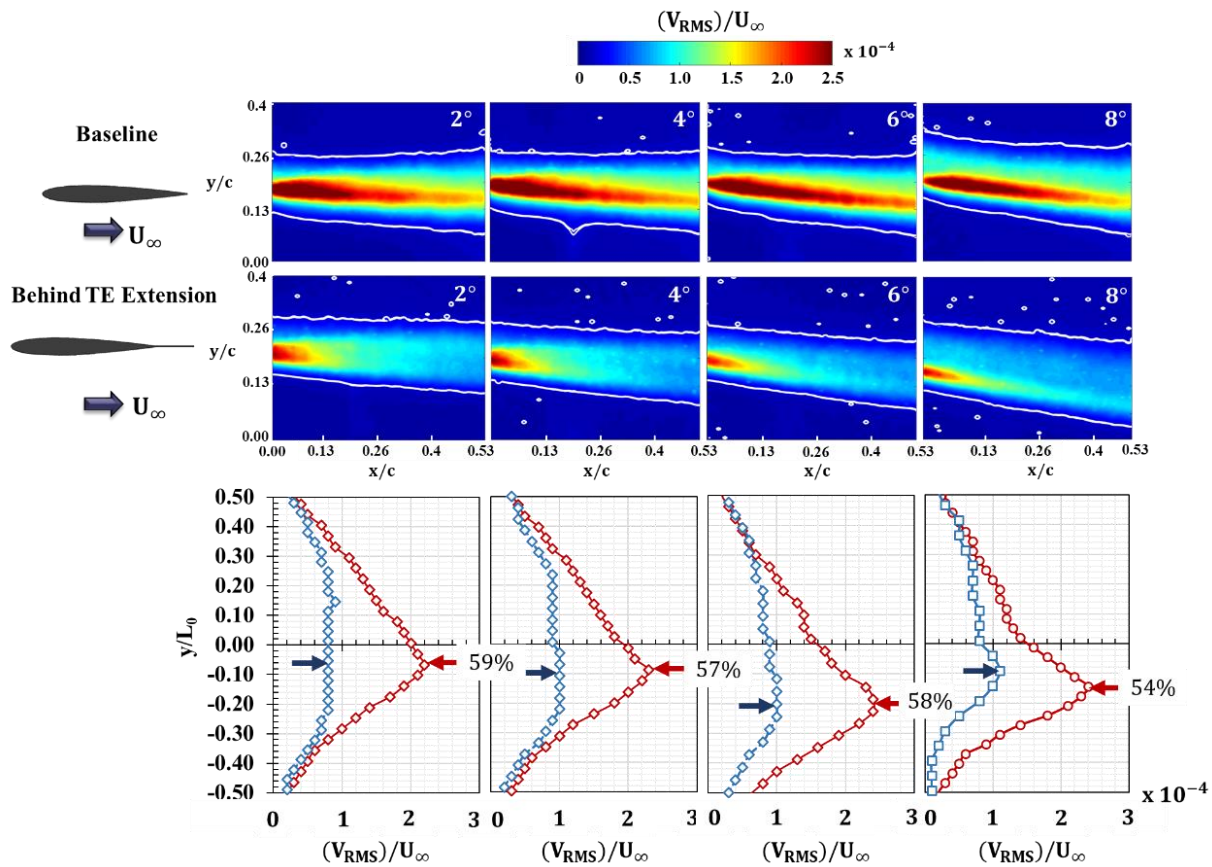


Figure 18 a) Streamwise V_{RMS} contours in FSL behind the trailing edge of the NACA 0012 wing and behind the TE extension b) V_{RMS} profiles at different angles of attack for both cases. Large decreases were observed in the V_{RMS} of the TE extension

4. Conclusion

The wake behind a NACA 0012 baseline semi-span wing was investigated with and without 0.25c rigid trailing edge (TE) extensions. Force and PIV experiments were conducted to analyze how the segmented TE extensions affected the vorticity and turbulent signatures in the wake. The prominent conclusions from the research are:

1. Adding the TE extensions had little effect on the coefficient of lift but had a minor impact on drag at lower angle of attack and measurable impact on drag at high angles of attack. With the segmented TE extensions, the total drag reduced by 8% at 8°.
2. Evidence for the cause of reduction in parasitic drag with TE extensions was supported by mean flow quantities such as mean velocity and normalized vorticity. Both parameters showed measurable and significant reductions when compared to the baseline especially in the vorticity case. The average reduction in vorticity is in the order of 40% at 8° angle of attack.
3. The reduction in vorticity behind TE extension was further supported by determining the coherent structures in the wake. The TE extensions reduced the length scales and the correlation between the wake and the upper surface boundary layer which indicates lower velocity and pressure fluctuations.
4. The lower pressure fluctuations can be supported by the changes observed in the Reynolds stress. On an average, the magnitude of the Reynolds stress was reduced by 40% on the upper surface and by 55% on the lower surface.
5. The reduction in fluctuations are further validated by determining U_{RMS} and V_{RMS} which showed an average decrease in the magnitude by 15% and 57% respectively.

These results provide evidence to consider segmented trailing edge extensions as a means to reduce turbulent fluctuations and vortex shedding in the wake of the wing without compromising on the lift production.

5. Future Work:

The work presented in this research is on-going, and sensitivity analysis on the aspect ratio of the trailing edge extensions are currently being explored. The results will be reported in a follow-up paper. The use of flexible segmented trailing edge is also under investigation along with studying the effects of the TE extensions on the wingtip vortex rollup process.

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